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Starting the Process

“Where shall I begin, please your Majesty?” he asked.

*“Begin at the beginning,” the King said gravely,
“and go on till you come to the end: then stop.”*

—LEWIS CARROLL

ALICE’S ADVENTURES IN WONDERLAND

Topics Covered in This Chapter

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Summary

Review Questions

Everything has a beginning, and the database design process is no different. Interestingly enough, you start the process by defining the end result. It is in the very first step of the database design process that you identify and declare the purpose of the database. You also define and declare a list of the tasks that your users can perform against the data in the database. Both of these items provide you with a focus and direction for developing a database, and they help ensure that your final database structure supports the stated purpose and tasks.

❖ **Special Note** The COVID-19 pandemic of 2020 has changed the way people conduct business and how we interact with each other. A far greater number of people are working remotely, especially from home. A lot of companies were already allowing their employees to work remotely prior to this, and many were using popular meeting platforms to conduct their meetings. But the pandemic just accelerated that idea on a much bigger scale. At the time of this writing, major companies, organizations, and government agencies are rethinking their work models, and many are likely to keep the idea of remote working in some form or fashion as a permanent part of their new way of doing business.

In light of these changes, I've adjusted my discussion of interviewing accordingly. My advice and guidelines now approach interviews from the perspective of meetings you're likely to conduct over meeting platforms such as Skype, Zoom, Microsoft Teams, or Google Meet. As always, I'll cover just the conceptual issues of the interviews; you'll have to refer to the help content or "How To" videos of the platform you're using to learn how to conduct the meeting in that platform.

Conducting Interviews

Interviews are an integral part of database design, and they play a key role during certain phases of the design process. Assuming that you work within some organization and need to design a database to support the work that you and your fellow employees perform, you should make certain that you conduct your interviews in the manner described in this book. This means that you'll interact with some of your fellow employees, management personnel, and the owner (depending on the size of the organization) throughout the design process. If you work for a small organization that employs only a handful of

people or are only creating a database for yourself, you'll conduct "self-interviews"; you'll still conduct the interviews described in this book, but you will act as the interviewer *and* the interviewee. *You* will be the one who provides the answers to the questions.

❖ **Note** Interviewing is a skill that you can learn with some amount of patience, diligence, and practice. You can use a variety of approaches and techniques to conduct an interview, and numerous academic papers, articles, and books are available on the subject. An in-depth discussion of this topic is beyond the scope of this book, but I've included several techniques and guidelines in this chapter that will help you conduct your interviews efficiently and effectively.

Interviews are important because they provide a valuable communication link between you (the developer) and the people for whom you're designing the database, help ensure the success of your design efforts, and provide critical information that can affect the design of the database structure. As you're working with table relationships, for example, you might find it difficult to determine the type of participation and degree of participation for a few relationships. The only way for you to determine the proper values for these relationship characteristics is to conduct an interview with the appropriate people in your organization. You can then use the information you gathered during the interview to set the relationship characteristics. You can use an interview as an information-gathering tool to gain new insights from participants regarding part of the database or to clarify facts that you don't understand. Note that you must always conduct each of the interviews incorporated within this design process, regardless of the type of database you're designing or the number of people involved. You will inevitably miss some piece of important information when you neglect or omit any of the interviews, and this could adversely affect the final structure of your database.

❖ **Note** Throughout the remaining chapters, I use open-ended questions for all interviews that are part of the concept or technique under discussion. You can use these questions as a guide for formulating your own questions for a given interview.

Always establish guidelines for your interviews before you conduct them. This helps ensure that you conduct your interviews in a consistent manner and that they are always (or usually) successful. The following sections offer some guidelines you can establish for the participants and for yourself.

Participant Guidelines

- *Make the participants aware of your intentions.* Many people are wary of interviews. They don't like to be "put on the spot," and they don't want to be asked what they feel might be "trick" questions. Let each person know the subject you want to discuss, who else is likely to participate in the discussion, the time you want to start the session, and whether this interview is part of an ongoing series of interviews. Everyone in a given interview session is more likely to engage in the conversation at hand and be quite responsive to your questions if they know how you're going to conduct the session and what you expect of them. Above all, reassure them that the interview is not a disguised assessment of their performance; you want to make certain they feel comfortable talking to you openly and without reservation. This will go a long way toward building a foundation of trust between you and the participants.
- *Let the participants know that you appreciate their participation in the interview and that their responses to the interview questions*

are valuable to the overall design project. Earlier experiences are likely to make some people believe that whatever input they provide at work goes unnoticed and unappreciated. Even when their input did make a significant impact on a specific project, rarely did they get so much as a “Thank you.” In light of this, there’s no real motivation for them to participate in your interview. Many, if not all, of your participants are likely to start out with this attitude, but you can really increase their motivation by letting them know that you sincerely and honestly appreciate their participation and are very interested in their responses. Assure them that their feedback is truly valuable to the design process and that in many cases their responses can substantiate and validate decisions made throughout the design process. The participants will be more apt to help you in any way they can if you make yourself credible by being genuinely sincere; your job will be much easier and everyone will participate voluntarily and enthusiastically. It’s also very effective to show, on a second interview, how you have already used participants’ earlier contributions.

- *Make sure everyone understands that you are the official arbitrator if and when a dispute arises.* It’s inevitable that minor disputes will arise during an interview and that there will be some amount of tension until such disputes are resolved. You can avoid this situation by arbitrating these disputes yourself. As the database developer, you’re in the best position to do this because you have an objective viewpoint and can see both sides of an issue. Additionally, the decision you make will always be in the best interests of the database structure. Always remember that disputes dealing with something other than the database structure can and should be referred to a more appropriate authority, if one exists.

Interviewer Guidelines (These Are for You)

- *Choose an appropriate meeting platform to conduct your meeting.* A few popular meeting platforms that you can use to conduct your meeting include Skype, Zoom, WebEx, Microsoft Teams, and Google Meet. If you're in a large organization, you'll most likely need to use the one that is the approved standard. The advantage here is that most people are likely to have used it at some point and will be familiar with how it works. If you're in a smaller organization or have a small business, choose the platform that is most likely to be the most intuitive and easiest to use for your intended audience. This is an especially good idea if you have folks who might be totally unfamiliar with or uncomfortable using these types of platforms.
- *Set a reasonable, practical limit for the number of people participating in each interview.* Limiting the number of participants promotes a more relaxed setting and makes it easier for you to encourage everyone to participate. One problem you'll find in conducting an interview with a large number of people is that the intimidation level of some of the participants will rise in direct proportion to the number of participants taking part in the interview as a whole. Some people are just afraid of looking ignorant or incompetent in front of their colleagues, regardless of whether there's truly any justification for such feelings. As such, you do have a very good reason to restrict the number of participants in the interview.
- *Conduct separate interviews for users and management.* Separating the two groups is a good idea for a variety of reasons, including the "fear factor" I noted earlier. Primarily, you want to separate them because each group has a different perspective on the organization as a whole and how the organization uses its data on a daily basis. Conducting separate interviews for each group allows you to leverage their unique perspectives to your

advantage as you work through the database design process. Another reason for keeping the interviews separate is to eliminate the conflicts that can arise when these groups disagree about certain aspects of the organization. It's quite common for there to be a lack of communication between them, and the odds are 50/50 that the interview will bring this problem to the surface. This may impel them to establish better lines of communication, or it may exacerbate the problem further. In any case, this communication problem can complicate and extend your interview and diffuse its results. Use your knowledge of the organization to help you judge whether to keep the interviews separate. If you need to conduct an interview with both groups at the same time, do so *intentionally*, with a specific purpose in mind, and be prepared for distractions.

- *Prepare your questions prior to the interview.* You can conduct an interview rather easily if you have a set of prepared questions. (Coming up with questions off the top of your head is rarely a good idea, even if you're an experienced interviewer and are highly skilled at producing ad hoc questions.) Having a prepared list of questions allows you to provide a focus and direction for the interview, and it provides the participants with a continuity of thought. Your interview will flow more smoothly and will be more productive when your questions move easily from topic to topic.

As you prepare your list of interview questions, make sure you use open-ended questions. For example, "Did you feel our service was (a) poor, (b) average, or (c) good?" is a closed question. A closed question isn't particularly useful because it supplies its own set of responses and does not allow an interviewee to provide an objective opinion or elaborate answer. On the other hand, an open-ended question such as "How do you feel about our service?" is far more useful because it allows the interviewee to answer the question in a variety of ways. There are times when

you may need to use closed questions, but it's better to use them intentionally, sparingly, and with a specific purpose in mind.

- *If you're not very good at taking notes, either assign that task to a dependable transcriber for each interview or inform the group ahead of time that you're going to record the interview for reference purposes.* You conduct interviews to gather specific information about the organization, so it's important that you establish a reasonably detailed record of each interview. If you find it difficult to conduct an interview and take notes at the same time, you should enlist one of the participants as your assistant to take notes for you. (This is one good way to encourage participation from people who are normally quiet or reserved.) Choose your assistant carefully because the notes may suffer if that person is at all distracted by the proceedings. Another option you have available is to record the interview proceedings. This might prove to be a better way to handle your notes because the recorder will capture the interview more accurately, and you'll be able to determine exactly who provided you with a given piece of information. Be sure you notify all participants prior to the interview if you do decide to record it for reference purposes. Privacy or confidentiality issues may be at stake, and you don't want to get yourself or anyone else into any kind of trouble.
- *Give everyone your equal and undivided attention.* This is a crucial point for you to remember—you must pay complete attention to the person who is speaking and do so sincerely. If you give participants the impression that you're bored, uninterested, or preoccupied, they will immediately reduce their level of participation within the interview. On the other hand, they will probably participate quite enthusiastically if they see that you are interested in what they're saying and have your undivided attention.

I'm sure you know that there will be times when a participant responds to your questions with vague or incomplete answers.

They may respond this way for several reasons. It may be that they don't quite know how to express the ideas they want to convey or that they're not at liberty to divulge certain information. It could also be that they're just not comfortable talking about themselves and what they do or that they are suspicious of you for some reason. You'll just have to be patient and make them feel at ease so that they will provide you with the information you need. For example, you could try to state your best approximation of what been said thus far and ask if it is what they meant to say.

- *Keep the pace of the interview moving.* You've probably attended meetings during which a particular point was belabored or much time was spent trying to extract information from a reluctant participant. You can prevent this from happening during your interviews by setting personal limits on the time you'll allow for a question to be answered and the time you'll spend on a specific topic. Don't inform the participants about this limit; instead, indicate that you'll table the point for now so that the meeting can proceed. Be sure you get in touch with the owner of the database soon after the meeting so that you can come to a final conclusion and resolution to the issue.
- *Always maintain control of the interview.* This is the single most important guideline for every interview you conduct. Inevitably, something goes wrong the moment you lose control of the interview. For instance, say you have a situation where one of the participants begins to change the focus of the interview by discussing issues that have little or no relevance to the topics on your agenda. You'll certainly lose control of the interview unless you do something to redirect the discussion. Regaining control of the interview will be easy for you to do in some cases, but in others you'll just have to declare your portion of the interview "complete" and let the participants carry on with their discussion.

You'll also occasionally encounter a participant who may want to dominate the interview and answer every one of your questions. When this happens, diplomatically and politely inform them that it is your job (and duty) to obtain feedback from all the participants so that you can make a complete assessment of the organization's overall information requirements. If this doesn't rectify the problem, you always have the option of refraining from including that participant in future discussions. You can avoid situations like these as long as you maintain control of the interview. Interviews are an integral part of the design process, and I provide examples of them throughout the next several chapters. You'll find sample dialogue that illustrates typical interview scenarios and examples of questions you might use during a given interview. (The sample questions always relate to the type of interview you're currently conducting.)

❖ **Note** The purpose of an interview example is to illustrate the *techniques* you use to conduct a specific type of interview, and I've kept the dialogue relatively simple for this reason. Use the dialogue as a means of coming up with good ideas for the types of conversations you conduct in the interview.

One final point: Keep in mind that the guidelines I've presented in this section are merely *recommendations*. I suspect that you won't be able to apply all of these guidelines to every interview you conduct or even apply them to the extent to which I've described. I would, however, expect you to apply them fully in an ideal situation. Yes, I know—you don't come across ideal situations all the time. Neither do I. But you can still make it your goal to meet as many of these guidelines as possible. In the end, the person who stands to gain the most is you.

Defining the Mission Statement

In the previous chapter, you learned that the *mission statement* declares the specific purpose of the database in general terms and that you define it at the beginning of the database design process. Furthermore, it provides you with a focus for your design efforts and keeps you from getting diverted and making the database structure unnecessarily large or complex.

The Well-Written Mission Statement

A good mission statement is succinct and to the point. Verbose statements have a tendency to be confusing, ambiguous, or vague; they do more to obscure the purpose of the database than to clarify it. Here is an example of a typical mission statement:

The purpose of the New Starz Talent Agency database is to maintain the data we generate and to supply information that supports the engagement services we provide to our clients and the management services we provide to our entertainers.

This mission statement is well defined and uncluttered by unnecessary statements or details. It is a very general statement, just as it should be. Think of a mission statement as the flame of a candle located at the end of a dark tunnel. The light produced by the flame guides you to the end of the tunnel, as long as you focus on it. In the same manner, the mission statement guides you to the end of the database design process. Guided by your mission statement, you can focus on designing a database structure that will support the declared purpose of the database.

A well-written mission statement is free of phrases or sentences that explicitly describe *specific tasks*. If your mission statement contains these types of phrases or sentences, remove them and rewrite the statement. Be sure to keep the discarded phrases handy, though,

because you may be able to use them to formulate mission objectives. (You'll learn about mission objectives in the next section.) Here's an example of a poorly worded mission statement:

The purpose of the Whatcom County Hearing Examiner's database is to keep track of applications for land use, maintain data on applicants, keep a record of all hearings, keep a record of all decisions, keep a record of all appeals, maintain data on department employees, and maintain data for general office use.

It should be immediately apparent that a few things are wrong with this mission statement.

- *It's somewhat verbose.* Remember that the ideal mission statement should be succinct and to the point.
- *The specific purpose of the database is unclear.* This mission statement is written in such a way that ascertaining the specific purpose of the database is difficult.
- *It describes several specific tasks.* Two issues arise when a mission statement is written in this manner. First, the description of the tasks does nothing to define the specific purpose of the database. Second, the statement somehow appears to be incomplete. It raises the question, "Are there any tasks we've forgotten to include in the mission statement?"

You can fix this mission statement by removing the references to specific tasks (be sure to save them for the next step) and rewriting the statement. Here is an example of one of the possible ways you could rewrite this mission statement:

The purpose of the Whatcom County Hearing Examiner's database is to maintain the data the examiner's office uses to make decisions on land-use requests submitted by citizens of Whatcom County.

Notice how the purpose of the database has become much clearer in this version. Also note that the statement is more succinct and doesn't

give the impression of being incomplete. You'll always have a clear focus during the database design process when you formulate your mission statements in this manner.

Composing a Mission Statement

The process of creating a mission statement involves conducting an interview with the owner or manager of the organization, learning about the organization, and determining the purpose of the new database.

You conduct the interview for this step with the owner of the organization or, if they direct, the appropriate staff member. Either will be able to help you define the statement because each has an overall understanding of the organization and a general comprehension of why the database is necessary in the first place. Besides helping you to define the mission statement, this interview will also provide a great deal of information about the organization itself. This information is valuable because you can use it later in the design process.

Encourage interview participants to discuss as many facets of the organization as they can, even if the discussion relates to issues that aren't directly relevant to the database. The idea here is for you to understand what the organization does and how it functions; the more you understand an organization, the better prepared you will be to design a database that will fulfill its needs. The organization's general need for a database will become clear to you after you have a better understanding of the organization itself. You can then translate this need into a mission statement.

Be sure to ask open-ended questions during the interview. In some cases, a good question can prompt the participant to state the purpose of the database without much effort. For example, say you posed the following question:

"How would you describe the purpose of your organization to a new client?"

This is a good open-ended question because it focuses on the issue and yet gives the participant the freedom to respond with what they feel is a complete answer. Furthermore, this type of question will typically generate a response that you can translate directly into a mission statement.

Now assume you received the following reply:

“We supply entertainment services to our clientele for any and all occasions. We take care of all the details for the engagement so that it is as worry-free for the client as possible.”

You can easily rewrite this type of response and turn it into a mission statement. When a response such as this one consists of two or more sentences or phrases, one of the sentences or phrases typically indicates the purpose of the database. For example, you can use the first sentence from the preceding reply to construct the mission statement. Here is one of several ways you could rewrite the reply:

The purpose of the All-Star Talent database is to maintain the data we use in support of the entertainment services we provide to our clientele.

The most important point to remember is that the mission statement should make sense to you (the database developer) and to those for whom you are designing the database. Different groups of people have different ways of phrasing statements, and the specific wording of the statement can depend greatly on industry-specific terminology. Your mission statement is complete when you have a sentence that describes the specific purpose of the database and that is understood and agreed upon by everyone concerned.

Here are a few sample questions that you can use to arrive at your mission statement:

How would you describe the purpose of your organization to a new client?

What would you say is the purpose of your organization?

What is the major function of your organization?

How would you describe what your organization does?

How would you define the single most important reason for the existence of your organization?

What is the main focus of your organization?

You may have noticed that some of these questions seem to be the same questions rewritten in a different manner. Keep in mind that the observation regarding the phrasing of mission statements also applies to the interview questions you'll use throughout the database design process. You can pose the *same question* to several people and receive different responses because each person may interpret the meaning of the question a little differently. In some cases, you may just get a long "I haven't had my first espresso yet" type of stare. Experiment with different types of phrasing and determine which type works best for you. Your method of constructing and posing questions may be different from someone else's, but it doesn't matter as long as you have a method that suits you.

EXAMPLE: DEFINING A MISSION STATEMENT

Let's see how you would define a Mission Statement for a small business known as "Mike's Bikes." It's a small bike shop located in a suburban shopping area among other small businesses. Before you can define the mission statement, you must conduct an interview with the owner, Mike, to gather information about his business. The interview may go something like this:

YOU: "Can you tell me why you believe you need a database?"

MIKE: "I think we need a database just to keep track of all our inventory. I'd also like to keep track of all our sales as well."

YOU: “I’m sure the database will address those issues. Now, what would you say is the single most important function of your business?”

MIKE: “To provide a wide array of bicycle products and bicycle-related services to our customers. We have a lot of great customers. And regular ones, too! They’re our biggest asset.”

(The interview continues until you’re finished asking all the questions on your list.)

After the interview, review the information you’ve gathered and define the mission statement. You can ascertain a few points from the previous dialogue with Mike, such as the fact that he’ll need to be able to track products, customers, and customer sales. But the most valuable point is provided by his reply to the second question. You can use the first sentence in that reply to formulate the mission statement. Taking into account some of the other points you’ve identified in the interview, you can rewrite Mike’s reply to create the following mission statement:

The purpose of the Mike’s Bikes database is to maintain the data we need to support our retail sales business and our customer service operations.

When you believe you have a good mission statement, review it with Mike and make sure that he understands and agrees with the declared purpose of the database. When you and Mike are satisfied with the mission statement, you can go on to the next step, which is to define the mission objectives.

Defining the Mission Objectives

To expand upon the overview in the previous chapter, *mission objectives* are statements that represent the *general* tasks supported by the data maintained in the database. Each mission objective represents a *single* task. These mission objectives provide information that you’ll

use throughout the database design process. For example, mission objectives help you define table structures, field specifications, relationship characteristics, and views. They also help you establish data integrity and define business rules. Finally, mission objectives guide your development efforts and ensure that your final database structure supports the mission statement.

Well-Written Mission Objectives

A well-written mission objective is a declarative sentence that clearly defines a general task and is free from unnecessary details. It is expressed in general terms, succinct and to the point, and unambiguous. Here are some examples of typical mission objectives:

Maintain complete patient address information.

Keep track of all customer sales.

Make sure an account representative is responsible for no more than 20 accounts at any given time.

Keep track of vehicle maintenance.

Produce employee phone directories.

These mission objectives are well defined and easy to understand. Each mission objective represents a single general task and defines the task clearly without unnecessary details. For example, the last mission objective in the list states that employee directories need to be produced, but it doesn't indicate *how* they are to be produced. Indicating how the employee directories will be produced is not necessary because that issue is part of the application development process. Remember that the purpose of a mission objective is to help define various structures within the database and to help guide the overall direction of the database's development.

If a mission objective represents more than one general task, you should decompose it into two or more mission objectives. Here is an example of a poorly written mission objective:

We need to keep track of the entertainers we represent and the type of entertainment they provide, as well as the engagements that we book for them.

There are two problems with this mission objective.

1. *It defines more than a single general task.* It is clear that there are two tasks represented in this statement—keeping track of entertainers and keeping track of engagements.
2. *It contains unnecessary detail.* Referring to the entertainer’s “type of entertainment” in this mission objective is unnecessary. The phrase *type of entertainment* either refers to a distinct characteristic of an entertainer or represents a new task that should be declared as a mission objective. If it refers to a distinct characteristic of an entertainer, it should be removed from the statement; otherwise, it should be used as the basis for a new mission objective.

You can fix this mission objective by removing the unnecessary detail and rewriting it as two mission objectives. (Keep the details you discard on a separate list; they may be useful later in the design process.) Here is an example of one possible revision:

Maintain complete entertainer information.

Keep track of all the engagements we book.

Notice that each mission objective now clearly defines a single general task and is easy to understand as well. Mission objectives such as these are easy to use as you design the database.

Composing Mission Objectives

Defining mission objectives is a process that involves conducting interviews with users and management and then writing appropriate mission objectives based on the information gathered from the interviews.

The purpose of the interview is to determine what types of general tasks need to be supported by the data in the database. You accomplish this by asking the participants open-ended questions and allowing them to elaborate on their replies as necessary. The mission statement and mission objectives interviews are the easiest ones you'll conduct during the design process because everyone is usually enthusiastic about participating. (In my experience, at least.) It's fairly easy to get people to discuss what they do on a daily basis and to give their perspective on the function of the organization. This is also one of the few interviews you'll conduct with both users and management; there should be a lot of common ground between the two groups due to the general nature of the interview.

One very important point to remember is that *the interviews you conduct here involve very general discussions*. The discussions are more conceptual than analytical; your intent here is not to analyze the current database or database application, but to get an overall idea of the general tasks the database should support. Keep in mind that one of the purposes of the mission objectives is to help guide the development of the database structure.

As you conduct the interview, be sure, once again, to ask open-ended questions. Remember that open-ended questions are apt to elicit better responses from your participants. Ask the participants questions regarding their daily work, how the organization functions, and what type of issues they believe need to be addressed by the database. Encourage them to discuss as many facets of their work and the organization as they possibly can. As they reply, try to record each response as a declarative sentence. You'll find it is much easier to

transform a sentence into a mission objective if you can do this. Here are just a few examples of the types of questions you could pose during the interview:

What kind of work do you perform on a daily basis?

How would you define your job description?

What kind of data do you work with?

What types of reports do you generate?

What types of things do you keep track of?

What types of services does your organization provide?

How would you describe the type of work you do?

All of these questions are likely to evoke a good, lengthy response from the participant. One of the advantages of questions like these is that they provide the opportunity for you to ask follow-up questions. For example, say you received the following response to the last question in the list:

“First, I try to determine the general problem with the vehicle. Then I fill out a work order and note my assessment of the problem. Finally, I send the vehicle to the next available service team.”

You’ll immediately notice that it’s a lengthy response, which is fine. You should also note that you could easily ask a follow-up question, such as the following:

“Is there any type of customer information incorporated within the procedure you just described?”

Even if the reply is “No,” the question is still open-ended enough for the participant to elaborate further on his original response. This type of follow-up question could also jar his memory and cause him to relay other information, which may be related to the subject of the original response.

Here is a set of mission objectives that you could derive from the participant's original response:

Maintain information on customer vehicles.

Keep track of work orders.

Maintain information on our service teams.

Maintain information on our mechanics.

Maintain information on our customers.

Three of these objectives are derived directly from the response. They're easy for you to determine because their subjects are explicitly stated in the response itself. The last two mission objectives are derived from *assumptions* based on the response. This is a technique (which you can think of as “reading between the lines”) that experienced database designers use quite often, and it is one that you should use when you're defining mission objectives. The technique relies on your ability to determine what information a response conveys *implicitly*, as well as what it conveys *explicitly*. So pay attention. Listen for implications. Without good assumptions, your overall set of mission objectives could be incomplete.

Review the following response and determine whether there is implicit information hidden within the response itself:

“I book entertainment for our clientele, which consists of commercial and noncommercial clients. Our noncommercial clients are typically individuals or small groups who book weddings, birthdays, anniversaries, and the like. Our commercial clients, on the other hand, consist of businesses such as nightclubs and corporations. The nightclubs book entertainment in six-week slots; the corporations book things such as corporate parties, product rollouts, and various types of promotional functions.”

Aside from the explicit information that this response conveys, there are at least two pieces of implicit information that you can uncover in

this response. The first piece of implicit information concerns the need to maintain information on the entertainers booked for the engagements. An agent needs to know things such as the entertainer's name, phone number, mailing address, availability, and whether he will travel to out-of-town locations. The second piece of implicit information concerns the need to maintain information on the engagements themselves. An agent must know all the details concerning the engagement to ensure that the engagement runs smoothly.

Now that you know how important it is to look for implicit information, keep it in mind when you're defining mission objectives.

Here's the bottom line regarding mission objectives: Make sure that your mission objectives are both properly defined and well defined, that each objective makes sense to you and to those for whom you are designing the database, and that you look for any implicit information hidden within every participant's response.

EXAMPLE: DEFINING MISSION OBJECTIVES

Let's work with Mike again and interview him so that he can help you define the mission objectives for his database. Here's a partial transcript of the interview with Mike. Once again, you're conducting the interview.

YOU: "Can you give me an idea of the things you'd like to track in the database?"

MIKE: "Oh sure, that's pretty easy. I want to keep track of our inventory, our customers, and our sales."

YOU: "Is there anything else that you can think of that is related to these subjects?"

MIKE: "Well, I guess if we're going to keep track of our inventory, we should know who our suppliers are."

YOU: "What about the sales reps involved in each sale?"

MIKE: “Oh yeah, we should definitely keep information about our employees. If nothing else, it’s a good idea to do this from a human resources point of view. At least, that’s what my wife tells me!”

(The interview continues until you’ve finished asking all the questions on his list.)

When the interviews are complete, review all the information you’ve gathered and define the appropriate mission objectives. Be sure to keep the bottom line in mind as you define them. Here are a few possible mission objectives for the Mike’s Bikes database.

Maintain complete inventory information.

Maintain complete customer information.

Track all customer sales.

Maintain complete supplier information.

Maintain complete employee information.

After you’ve compiled a list of mission objectives, review them with Mike and his staff. When they are satisfied that they understand the mission objectives and that the list is relatively complete, commit the list to a document in your favorite application program and save it for later use.

Summary

This chapter opened with a discussion of the *interview process*. You learned why interviews are an important part of the database design process and why it’s important to learn how to conduct an interview properly. You now know the difference between an *open-ended* question and a *closed* question, as well as when to use each kind of question. We ended this discussion by reviewing a set of interview guidelines,

and you learned that you should use them to help you ensure that the interviews are productive and successful.

The *mission statement* was our next topic of discussion. We expanded upon the information in Chapter 4, “Conceptual Overview,” by looking at how the mission statement states the specific purpose of the database. You now know that the process involves conducting interviews and learning about the organization, then formulating the mission statement from the information you gathered during these steps. We defined the characteristics of a good mission statement, and you learned that a well-defined mission statement establishes a clear focus for your design efforts.

Next, we discussed *mission objectives*, and we expanded upon the Chapter 4 overview once again. As you now know, mission objectives represent the tasks performed against the data in the database, and you define them after the mission statement. We then explored how to define a mission objective. Here, you learned that you conduct interviews with users and management and that the information you gather from these interviews provides the basis for each mission objective. We also discussed the characteristics of a well-written mission objective, and you learned that a clearly defined mission objective will help you define various structures within the database.

Review Questions

1. Why are interviews important?
2. What problem can arise when you conduct an interview with a large number of people?
3. What is the primary reason for conducting separate interviews with users and management?

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4. True or False: You'll commonly use closed questions in your interviews.
 5. What kind of responses should you try to evoke from the interview participants?
 6. What is the single most important guideline for every interview you conduct?
 7. What is a *mission statement*?
 8. State two characteristics of a well-written mission statement.
 9. True or False: You need not learn about the organization to compose a mission statement.
 10. When is your mission statement complete?
 11. What is a *mission objective*?
 12. State two characteristics of a well-written mission objective.
 13. True or False: You should interview users and management to help you define mission objectives.
 14. How does the staff's daily work relate to the mission objectives?
 15. True or False: A mission objective can describe more than one task.
 16. State two ways that a mission objective can be derived from a response.
 17. When is a mission objective complete?